PHISHING EMAIL DETECTION & AWARENESS REPORT

Cyber Security Task 02
Future Interns

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Introduction

Phishing is one of the most common cyber attacks used by attackers to steal sensitive information from users.

Attackers send fake emails pretending to be trusted organizations such as banks, companies, or service providers. These emails contain malicious links or instructions that trick users into revealing passwords, OTPs, or personal data.

This report analyzes phishing email samples, identifies phishing indicators, classifies email risks, and provides awareness guidelines to help users recognize and avoid phishing attacks.

The objective of this task is to analyze phishing email samples, identify common phishing indicators, classify the risk level of emails, and create awareness guidelines to help users detect and prevent phishing attacks.

Tools Used

The following tools were used during the phishing email analysis:

- Public Phishing Email Samples
- Google Message Header Analyzer
- MXToolbox Email Header Analyzer
- Web Browser
- Microsoft Word
- GitHub (for documentation and submission)

Email Sample Analysis

From: security@secure-account-verify.com
To: user@example.com
Subject: ⚠️ Urgent: Your Account Will Be Locked
Date: 12 May 2026

Dear User,

We noticed suspicious activity on your account.
To avoid account suspension, please verify your details immediately.

👉 Verify Now:
http://secure-account-verify[.]com

Failure to verify within 24 hours will result in permanent account lock.

Regards,
Security Team

From: billing@billing-update-secure.net
To: customer@example.com
Subject: Payment Failed – Update Billing Details
Date: 14 May 2026

Dear Customer,

Your recent payment could not be processed successfully.

To avoid service interruption, please update your billing information immediately

Update Now:
http://billing-update-secure[.]net

Thank you,
Billing Department

From: support@bank-security-check.org
To: customer@example.com
Subject: Bank Alert: Unauthorized Login Detected
Date: 15 May 2026

Dear Valued Customer,

We detected an unauthorized login attempt on your bank account.

For your security, please confirm your identity using the link below:

http://bank-security-check[.]org

If you do not verify within 12 hours, your account will be suspended.

Bank Security Team

Email 1 Analysis – Account Verification Email

Subject: **Urgent**: Your Account Will Be Locked
Sender: security@secure-account-verify.com

INDICATOR	FOUND	EXPLANATION
Fake Sender Domain	**Yes**	Sender domain looks suspicious and not from a trusted organization
Urgent Language	**Yes**	Uses phrases like "verify immediately" and "within 24 hours"
Suspicious Link	**Yes**	Link uses unknown domain (secure-account-verify.com)
Generic Greeting	**Yes**	Uses "Dear User" instead of real name
Fear Tactics	**Yes**	Threatens account lock to create panic

Risk Level: PHISHING (High Risk)

Reason:
This email contains multiple phishing indicators including fake domain, urgency, and suspicious link.

Email 2 Analysis — Payment Failure Email

Subject: **Payment Failed** – Update Billing Details
Sender: billing@billing-update-secure.net

INDICATOR	FOUND	EXPLANATION
Fake Sender Domain	**Yes**	Domain does not belong to a known payment service
Urgent Language	**Yes**	Asks user to update details immediately
Suspicious Link	**Yes**	Unknown billing website link
Generic Greeting	**Yes**	Uses "Dear Customer"
Finanacial Threat	**Yes**	Mentions service interruption

Risk Level: PHISHING (High Risk)

Reason:
This email attempts to steal billing details using urgency and fake links.

Email 3 Analysis — Bank Security Email

Subject: Bank Alert: Unauthorized Login Detected
Sender: support@bank-security-check.org

INDICATOR	FOUND	EXPLANATION
Fake Sender Domain	**Yes**	Domain does not match real bank domains
Urgency	**Yes**	Claims unauthorized login
Suspicious Link	**Yes**	Link directs to unknown website
Generic Greeting	**Yes**	Uses "Dear Valued Customer"
Account Threat	**Yes**	Threatens suspension

Risk Level: PHISHING (High Risk)

Reason:
Email impersonates a bank and uses urgent security warning to trick users

How Phishing Attack Works

Phishing attacks work by sending fake emails that appear to be from trusted organizations such as banks, online services, or companies.

The attacker creates an email that contains urgent messages and malicious links. These links redirect users to fake websites that look like real ones.

When users enter sensitive information such as passwords, OTPs, or banking details on these fake websites, the information is captured by attackers.

Once attackers obtain user credentials, they can access accounts, steal money, or misuse personal information.

Prevention Guidelines

Users can prevent phishing attacks by following these security practices:

- Always verify the sender's email address before responding.
- Do not click on unknown or suspicious links.
- Hover over links to check the actual website URL.
- Avoid downloading attachments from unknown senders.
- Enable Two-Factor Authentication (2FA) on important accounts.
- Use updated antivirus and security software.
- Report suspicious emails to the IT or security team.
- Never share passwords, OTPs, or personal information through email.

Do's and Don'ts

DO'S	DON'TS
Verify sender email address	Click unknown links
Check link URLs before clicking	Share passwords or OTPs
Enable Two-Factor Authentication	Download unknown attachments
Report suspicious emails	Ignore warning signs
Keep antivirus updated	Trust urgent emails blindly

Conclusion

Phishing attacks are one of the most common cyber threats faced by individuals and organizations. These attacks rely on social engineering techniques to trick users into revealing sensitive information.

By understanding phishing indicators and following safe email practices, users can reduce the risk of falling victim to phishing attacks.

Security awareness and proper user training play an important role in preventing phishing attacks and protecting sensitive data.